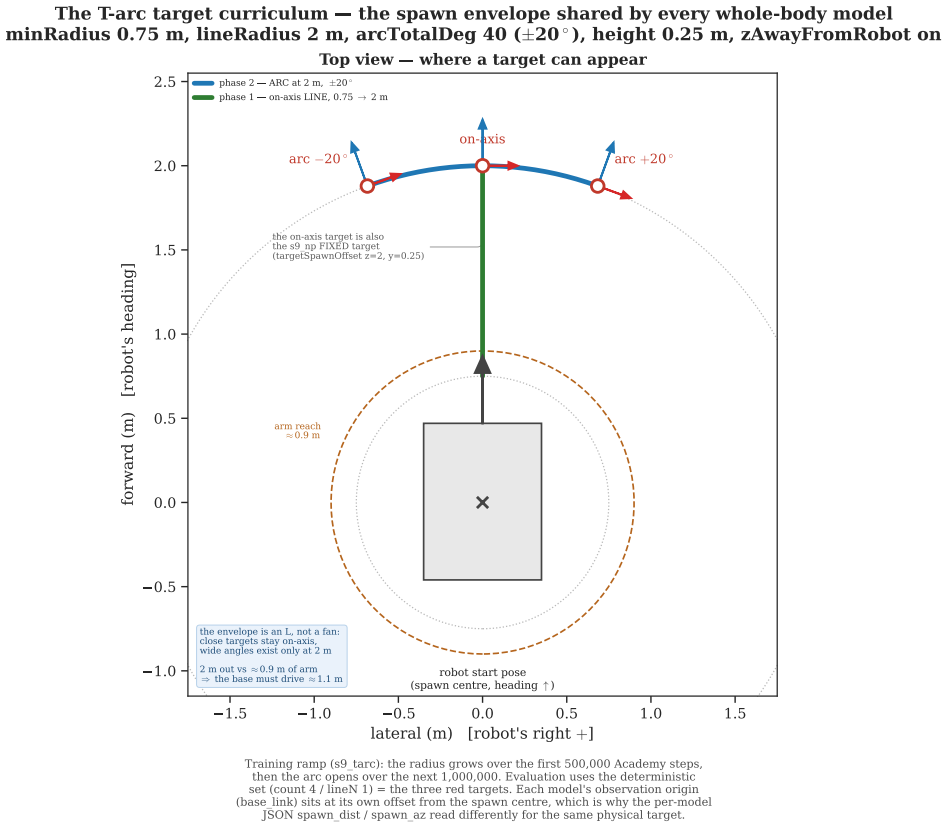
Common figures — Notes for the Defense

The task-level figures: they describe the problem, not a model, so they come BEFORE any per-model result and are shared by every whole-body package.

Figure 1 — The T-arc spawn envelope and the robot start pose



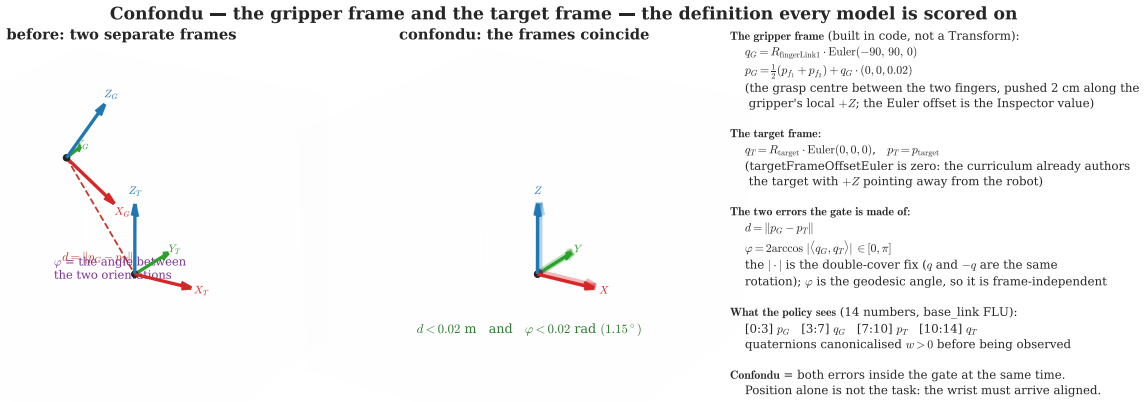
What it shows. Where a target can be created, relative to the robot's start pose, seen from above. The green segment is the on-axis line (0.75→2 m), the blue arc is the far arc at 2 m spanning ±20°, and the three red circles are the deterministic evaluation targets, drawn with their frame axes. The target always sits 0.25 m above the base.

How to read it. The envelope is an **L**, not a filled fan: the curriculum first pushes the target out along the axis, and only then opens the angle at the far radius, so a close target is never also strongly off-axis. The dashed circle carries the reason this is a whole-body problem at all: the target is at 2 m while the arm reaches about 0.9 m, so roughly 1.1 m has to be covered by driving the base.

Say this. "Every whole-body model is trained and evaluated on this same envelope — 2 m out, ±20°, 25 cm up — so the comparisons that follow are on identical targets."

If asked. *Why not sample the whole fan?* — because close-and-wide targets would be behind the robot's shoulder, a qualitatively different (and much harder) problem than the one this stage is about. *Why do the JSON spawn distances differ per model?* — they are measured from each robot's own `base_link`, which sits at a different offset from the spawn centre; the physical geometry is the same, and that was verified by fitting a circle to the logged target positions.

Figure 2 — What *confondu* means: the two frames and the two errors



What it shows. The gripper frame and the target frame as RGB triads (X red, Y green, Z blue — the same convention as the in-scene axis renderer), how each one is constructed from the serialized offsets, and the two error measures the success gate is made of.

How to read it. The left panel is a normal mid-episode state: the origins are d apart and the orientations differ by φ. The middle panel is the goal: the triads sit on top of each other. The gripper frame is *computed*, not a Transform in the scene — it is finger link 1's rotation times the Euler offset $(-90, 90, 0)$, and the origin is the grasp centre pushed 2 cm forward.

Say this. "The task is frame coincidence, not reaching: the gripper must arrive at the target's position *and* in the target's orientation — under 2 cm and 1.15° at the same time."

If asked. *Why the absolute value in φ?* — a quaternion and its negative are the same rotation; without $|\cdot|$ the metric reports the long way round (e.g. 330° instead of 30°). *Why is the gripper frame not just the finger's transform?* — the finger link's authored axes do not match the target's convention; the Euler offset is what makes the two frames comparable, and getting it wrong makes the gate unreachable without any error message.